

OSSP 7

Question:

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Solution:

```
ubuntu@ubuntu:~$ nano memory_layout.c
GNU nano 8.7.1 memory_layout.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;

void code_segment()
{
}

int main()
{
    static int static_var = 200;
    int stack_var = 300;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 400;

    printf("PID: %d\n", getpid());

    printf("Code Segment Address : %p\n", code_segment);
    printf("Global Variable Address: %p\n", &global_var);
    printf("Static Variable Address: %p\n", &static_var);
    printf("Heap Variable Address : %p\n", heap_var);
    printf("Stack Variable Address : %p\n", &stack_var);

    printf("\nProcess sleeping for 60 seconds...\n");
    sleep(60);

    free(heap_var);

    return 0;
}

ubuntu@ubuntu:~$ gcc memory_layout.c -o memory_layout
ubuntu@ubuntu:~$ ./memory_layout
PID: 6001

Code Segment Address : 0xb3d2ff4309e8
Global Variable Address: 0xb3d2ff450010
Static Variable Address: 0xb3d2ff450014
Heap Variable Address : 0xb3d31a5dc010
Stack Variable Address : 0xffffea87735c

Process sleeping for 60 seconds...
```

```
ubuntu@ubuntu:~$ cat /proc/3456/maps
```

```
6fb40100000-6fb40200000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
858a3650000-858a3660000 ---p 00000000 00:00 0 [anon:js-executable-memory]
858a3660000-858a3675000 r-xp 00000000 00:00 0 [anon:js-executable-memory]
858a3675000-858a3678000 r-xp 00000000 00:00 0 [anon:js-executable-memory]
858a3678000-858a367f000 r-xp 00000000 00:00 0 [anon:js-executable-memory]
858a367f000-858a3680000 r-xp 00000000 00:00 0 [anon:js-executable-memory]
858a3680000-858a3680000 ---p 00000000 00:00 0 [anon:js-executable-memory]
858a3680000-858a3780000 r-xp 00000000 00:00 0 [anon:js-executable-memory]
858a3780000-858a37a0000 r-xp 00000000 00:00 0 [anon:js-executable-memory]
858a37a0000-858a37a0000 r-xp 00000000 00:00 0 [anon:js-executable-memory]
858a37a0000-85923250000 ---p 00000000 00:00 0 [anon:js-executable-memory]
122c5f200000-122c5f300000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
13e64d100000-13e64d200000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
1507bb000000-1507bb100000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
196421c00000-196421d00000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
19be80500000-19be80600000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
2662a2b00000-2662a2c00000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
28f57c300000-28f57c400000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
2f6510000000-2f6510100000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
397b3d800000-397b3d900000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
3aabb3c100000-3aabb3c200000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
3c4463700000-3c4463800000 rw-p 00000000 00:00 0 [anon:js-gc-heap]
c54f3e350000-c54f3e355000 r-xp 00000000 00:21 3273 /usr/bin/gnome-shell
c54f3e36f000-c54f3e370000 r--p 0000f000 00:21 3273 /usr/bin/gnome-shell
c54f3e370000-c54f3e371000 rw-p 00010000 00:21 3273 /usr/bin/gnome-shell
c54f62e03000-c54f63000000 rw-p 00000000 00:00 0 [heap]
c54f63000000-c54f6ce00000 rw-p 00000000 00:00 0 [heap]
ffdda5d4a000-ffdda55a0000 rw-p 00000000 00:00 0
ffdda6da000-ffdda8000000 r--p 00000000 00:21 59390 /usr/share/fonts/opentype/noto/NotoSansCJK-Regular.ttc
ffdda8000000-ffddac000000 rw-s 00000000 00:01 3141 /memfd:pulseaudio (deleted)
ffddac000000-ffddac200000 rw-p 00000000 00:00 0
ffddac200000-ffddb0000000 ---p 00000000 00:00 0
ffddb0338000-ffddb15ce000 r--p 00000000 00:21 59390 /usr/share/fonts/opentype/noto/NotoSansCJK-Regular.ttc
ffddb15ce000-ffddb4200000 rw-p 00000000 00:00 0
ffddb4200000-ffddb8000000 ---p 00000000 00:00 0
ffddb81e0000-ffddb8420000 rw-s 00000000 00:01 2123 /memfd:wayland-cursor (deleted)
ffddb8638000-ffddb8878000 rw-s 00000000 00:01 2113 /memfd:wayland-cursor (deleted)
ffddb8878000-ffddb8ab8000 rw-s 00000000 00:01 1103 /memfd:wayland-cursor (deleted)
ffddb94c0000-ffddb9cd0000 rw-p 00000000 00:00 0
ffddb94e0000-ffddb9d10000 rw-p 00000000 00:00 0
ffddb9d10000-ffddb9d40000 r-xp 00000000 00:21 8737 /usr/lib/aarch64-linux-gnu/libmp3lame.so.0.0.0
ffddb9d40000-ffddb9d5f000 ---p 00038000 00:21 8737 /usr/lib/aarch64-linux-gnu/libmp3lame.so.0.0.0
ffddb9d5f000-ffddb9d60000 r--p 0003f000 00:21 8737 /usr/lib/aarch64-linux-gnu/libmp3lame.so.0.0.0
ffddb9d60000-ffddb9d61000 rw-p 00040000 00:21 8737 /usr/lib/aarch64-linux-gnu/libmp3lame.so.0.0.0
ffddb9d61000-ffddb9d71000 rw-p 00000000 00:00 0
ffddb9d70000-ffddb9d80000 r-xp 00000000 00:21 8743 /usr/lib/aarch64-linux-gnu/libmpg123.so.0.49.4
ffddb9d80000-ffddb9d9f000 ---p 00050000 00:21 8743 /usr/lib/aarch64-linux-gnu/libmpg123.so.0.49.4
ffddb9d9f000-ffddb9da0000 r--p 0005f000 00:21 8743 /usr/lib/aarch64-linux-gnu/libmpg123.so.0.49.4
ffddb9da0000-ffddb9dab000 rw-p 00060000 00:21 8743 /usr/lib/aarch64-linux-gnu/libmpg123.so.0.49.4
ffddb9dab000-ffddb9dab000 r-xp 00000000 00:21 8864 /usr/lib/aarch64-linux-gnu/libopus.so.0.11.1
ffddb9dab000-ffddb9daff000 ---p 003fb000 00:21 8864 /usr/lib/aarch64-linux-gnu/libopus.so.0.11.1
ffddb9daff000-ffddb9db0000 r--p 003fd000 00:21 8864 /usr/lib/aarch64-linux-gnu/libopus.so.0.11.1
ffddb9db0000-ffddb9db1000 rw-p 00400000 00:21 8864 /usr/lib/aarch64-linux-gnu/libopus.so.0.11.1
ffddb9db1000-ffddb9db9b000 r-xp 00000000 00:21 9306 /usr/lib/aarch64-linux-gnu/libvorbisenc.so.2.0.12
ffddb9db9b000-ffddb9db9a9000 ---p 0008b000 00:21 9306 /usr/lib/aarch64-linux-gnu/libvorbisenc.so.2.0.12
ffddb9db9a9000-ffddb9db9c0000 r--p 00099000 00:21 9306 /usr/lib/aarch64-linux-gnu/libvorbisenc.so.2.0.12
ffddb9db9c0000-ffddb9db9c1000 rw-p 000b0000 00:21 9306 /usr/lib/aarch64-linux-gnu/libvorbisenc.so.2.0.12
ffddb9db9c1000-ffddb9db11c000 r-xp 00000000 00:21 7635 /usr/lib/aarch64-linux-gnu/libFLAC.so.14.0.0
ffddb9db11c000-ffddb9db12f000 ---p 0004c000 00:21 7635 /usr/lib/aarch64-linux-gnu/libFLAC.so.14.0.0
ffddb9db12f000-ffddb9db130000 r--p 0004f000 00:21 7635 /usr/lib/aarch64-linux-gnu/libFLAC.so.14.0.0
ffddb9db130000-ffddb9db131000 rw-p 00050000 00:21 7635 /usr/lib/aarch64-linux-gnu/libFLAC.so.14.0.0
ffddb9db140000-ffddb9db154000 r-xp 00000000 00:21 7909 /usr/lib/aarch64-linux-gnu/libapparmor.so.1.29.4
ffddb9db154000-ffddb9db16f000 ---p 00014000 00:21 7909 /usr/lib/aarch64-linux-gnu/libapparmor.so.1.29.4
ffddb9db16f000-ffddb9db170000 r--p 0001f000 00:21 7909 /usr/lib/aarch64-linux-gnu/libapparmor.so.1.29.4
ffddb9db170000-ffddb9db171000 rw-p 00020000 00:21 7909 /usr/lib/aarch64-linux-gnu/libapparmor.so.1.29.4
ffddb9db180000-ffddb9db185000 r-xp 00000000 00:21 7926 /usr/lib/aarch64-linux-gnu/libasyncns.so.0.3.1
ffddb9db185000-ffddb9db19f000 ---p 00005000 00:21 7926 /usr/lib/aarch64-linux-gnu/libasyncns.so.0.3.1
ffddb9db19f000-ffddb9db1a0000 r--p 0000f000 00:21 7926 /usr/lib/aarch64-linux-gnu/libasyncns.so.0.3.1
ffddb9db1a0000-ffddb9db1a1000 rw-p 00010000 00:21 7926 /usr/lib/aarch64-linux-gnu/libasyncns.so.0.3.1
ffddb9db1b0000-ffddb9db23000 r-xp 00000000 00:21 9108 /usr/lib/aarch64-linux-gnu/libsndfile.so.1.0.37
ffddb9db230000-ffddb9db24d000 ---p 00083000 00:21 9108 /usr/lib/aarch64-linux-gnu/libsndfile.so.1.0.37
ffdrth74d000-ffdrth750000 r--p 0008d000 00:21 9108 /usr/lib/aarch64-linux-gnu/libsndfile.so.1.0.37
```

```

ffddc1f2d000-ffddc2035000 r--p 00000000 00:21 59755 /usr/share/fonts/truetype/ubuntu/UbuntuSans[wdth,wght].ttf
ffddc2035000-ffddc2045000 rw-p 00000000 00:00 0
ffddc2860000-ffddc2867000 r-xp 00000000 00:21 11051 /usr/lib/aarch64-linux-gnu/pipewire-0.3/libpipewire-module-rt.so
ffddc2867000-ffddc287f000 --p 00007000 00:21 11051 /usr/lib/aarch64-linux-gnu/pipewire-0.3/libpipewire-module-rt.so
ffddc287f000-ffddc2880000 r--p 0000f000 00:21 11051 /usr/lib/aarch64-linux-gnu/pipewire-0.3/libpipewire-module-rt.so
ffddc2880000-ffddc2881000 rw-p 00010000 00:21 11051 /usr/lib/aarch64-linux-gnu/pipewire-0.3/libpipewire-module-rt.so
ffddc2890000-ffddc2895000 r-xp 00000000 00:21 11586 /usr/lib/aarch64-linux-gnu/spa-0.2/support/libspa-dbus.so
ffddc2895000-ffddc28af000 --p 00005000 00:21 11586 /usr/lib/aarch64-linux-gnu/spa-0.2/support/libspa-dbus.so
ffddc28af000-ffddc28b0000 r--p 0000f000 00:21 11586 /usr/lib/aarch64-linux-gnu/spa-0.2/support/libspa-dbus.so
ffddc28b0000-ffddc28b1000 rw-p 00010000 00:21 11586 /usr/lib/aarch64-linux-gnu/spa-0.2/support/libspa-dbus.so
ffddc28c0000-ffddc2924000 r-xp 00000000 00:21 9104 /usr/lib/aarch64-linux-gnu/libsnapd-glib-2.so.1.0.0
ffddc2924000-ffddc293e000 --p 00064000 00:21 9104 /usr/lib/aarch64-linux-gnu/libsnapd-glib-2.so.1.0.0
ffddc293e000-ffddc2940000 r--p 0006e000 00:21 9104 /usr/lib/aarch64-linux-gnu/libsnapd-glib-2.so.1.0.0
ffddc2940000-ffddc2941000 rw-p 00070000 00:21 9104 /usr/lib/aarch64-linux-gnu/libsnapd-glib-2.so.1.0.0
ffddc295f000-ffddc2970000 r--p 00000000 00:21 7050 /usr/lib/aarch64-linux-gnu/girepository-1.0/Snapd-2.typelib
ffddc2970000-ffddc2987000 r-xp 00000000 00:21 6985 /usr/lib/aarch64-linux-gnu/gio/modules/libgioremove-volume-monitor.so
ffddc2987000-ffddc299d000 --p 00017000 00:21 6985 /usr/lib/aarch64-linux-gnu/gio/modules/libgioremove-volume-monitor.so
ffddc299d000-ffddc29a0000 r--p 0001d000 00:21 6985 /usr/lib/aarch64-linux-gnu/gio/modules/libgioremove-volume-monitor.so
ffddc29a0000-ffddc29a1000 rw-p 00020000 00:21 6985 /usr/lib/aarch64-linux-gnu/gio/modules/libgioremove-volume-monitor.so
ffddc2a00000-ffddc3460000 rw-p 00000000 00:00 0
ffddc3460000-ffddc37f0000 rw-s 00000000 00:01 131 /memfd:mesa-shared (deleted)
ffddc37f0000-ffddc4000000 rw-p 00000000 00:00 0
ffddc4000000-ffddc4200000 rw-p 00000000 00:00 0
ffddc4200000-ffddc8000000 --p 00000000 00:00 0
ffddc8000000-ffddc8200000 rw-p 00000000 00:00 0
ffddc8200000-ffddcc000000 --p 00000000 00:00 0
ffddcc000000-ffddcc200000 rw-p 00000000 00:00 0
ffddcc200000-ffddd0000000 --p 00000000 00:00 0
ffddd0000000-ffddd0200000 rw-p 00000000 00:00 0
ffddd0200000-ffddd4000000 --p 00000000 00:00 0
ffddd4000000-ffddd4200000 rw-p 00000000 00:00 0
ffddd4200000-ffddd8000000 --p 00000000 00:00 0
ffddd8000000-ffddd8200000 rw-p 00000000 00:00 0
ffddd8200000-ffdddc000000 --p 00000000 00:00 0
ffdddc000000-ffdddc200000 rw-p 00000000 00:00 0
ffdddc200000-ffdde0000000 --p 00000000 00:00 0
ffdde0000000-ffdde0200000 rw-p 00000000 00:00 0
ffdde0200000-ffdde4000000 --p 00000000 00:00 0
ffdde4000000-ffdde4200000 rw-p 00000000 00:00 0
ffdde4200000-ffdde4400000 --p 00000000 00:00 0

```

```
ubuntu@ubuntu:~$ cat /proc/3456/status
```

```

Name:   Mutter Input Th
Unmask: 0002
State:  S (sleeping)
Tgid:   3360
Ngid:   0
Pid:    3456
PPid:   2833
TracerPid: 0
Uid:    1000    1000    1000    1000
Gid:    1000    1000    1000    1000
FDSize: 256
Groups: 4 24 27 30 46 100 111 115 1000
NSTgid: 3360
NSpid:  3456
NSpgid: 3360
NSSid:  3360
Kthreid: 0
VmPeak: 5479544 kB
VmSize: 5081776 kB
VmLck:  0 kB
VmPin:  0 kB
VmHWM:  472376 kB
VmRSS:  472208 kB
RssAnon: 305520 kB
RssFile: 147660 kB
RssShmem: 19028 kB
VmData: 506696 kB
VmStk:  132 kB
VmExe:  20 kB
VmLib:  234720 kB
VmPTE:  1624 kB
VmSwap: 0 kB
HugetlbPages: 0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xfffffffffffff
Threads: 26

```

```
VmLck: 0 kB
VmPin: 0 kB
VmHWM: 472376 kB
VmRSS: 472208 kB
RssAnon: 305520 kB
RssFile: 147660 kB
RssShmem: 19028 kB
VmData: 506696 kB
VmStk: 132 kB
VmExe: 20 kB
VmLib: 234720 kB
VmPTE: 1624 kB
VmSwap: 0 kB
HugetlbPages: 0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xfffffffffffff
Threads: 26
SigQ: 0/13689
SigPnd: 0000000000000000
ShdPnd: 0000000000000000
SigBlk: 0000000000000000
SigIgn: 0000000010010000
SigCgt: 00000001000044fa
CapInh: 0000000000000000
CapPrm: 0000000000000000
CapEff: 0000000000000000
CapBnd: 000001ffffffffffff
CapAmb: 0000000000000000
NoNewPrivs: 0
Seccomp: 0
Seccomp_filters: 0
Speculation_Store_Bypass: vulnerable
SpeculationIndirectBranch: unknown
Cpus_allowed: f
Cpus_allowed_list: 0-3
Mems_allowed: 00000000,00000001
Mems_allowed_list: 0
voluntary_ctxt_switches: 2136
nonvoluntary_ctxt_switches: 38
```

```
ubuntu@ubuntu:~$ pmap -x 3456
```

```
3360: /usr/bin/gnome-shell --mode=ubuntu
```

Address	Kbytes	RSS	Dirty	Mode	Mapping
000006fb40100000	1024	880	880	rw---	[anon]
00000858a3650000	64	0	0	-----	[anon]
00000858a3660000	52	52	52	r-x--	[anon]
00000858a3675000	12	0	0	r-x--	[anon]
00000858a3678000	28	28	28	r-x--	[anon]
00000858a367f000	36	0	0	r-x--	[anon]
00000858a3688000	768	0	0	-----	[anon]
00000858a3748000	120	120	120	r-x--	[anon]
00000858a3766000	8	0	0	r-x--	[anon]
00000858a3768000	192	0	0	-----	[anon]
00000858a3798000	64	64	64	r-x--	[anon]
00000858a37a8000	2091712	0	0	-----	[anon]
0000122c5f200000	1024	940	940	rw---	[anon]
000013e64d100000	1024	384	384	rw---	[anon]
00001507bb000000	1024	732	732	rw---	[anon]
0000196421c00000	1024	820	820	rw---	[anon]
000019be80500000	1024	888	888	rw---	[anon]
00002662a2b00000	1024	980	980	rw---	[anon]
000028f57c300000	1024	744	744	rw---	[anon]
00002f6510000000	1024	104	104	rw---	[anon]
0000397b3d800000	1024	132	132	rw---	[anon]
00003aab3c100000	1024	516	516	rw---	[anon]
00003c4463700000	1024	256	256	rw---	[anon]
0000c54f3e350000	20	20	0	r-x--	gnome-shell
0000c54f3e36f000	4	4	4	r----	gnome-shell
0000c54f3e370000	4	4	4	rw---	gnome-shell
0000c54f62e03000	2036	1968	1968	rw---	[anon]
0000c54f63000000	165888	161792	161792	rw---	[anon]
0000ffdda5d4a000	8256	92	92	rw---	[anon]
0000ffdda6d6a000	19032	992	0	r----	NotoSansCJK-Regular.ttc
0000ffdda8000000	65536	0	0	rw-s-	memfd:pulseaudio (deleted)
0000ffddac000000	2048	2048	2048	rw---	[anon]
0000ffddac200000	63488	0	0	-----	[anon]
0000ffddb0338000	19032	704	0	r----	NotoSansCJK-Regular.ttc
0000ffddb15ce000	45256	45256	45256	rw---	[anon]
0000ffddb4200000	63488	0	0	-----	[anon]

```

0000fddb1000 224 64 0 r-x-- libmp3lame.so.0.0.0
0000fddb14000 92 0 0 ---- libmp3lame.so.0.0.0
0000fddb15f000 4 4 4 r---- libmp3lame.so.0.0.0
0000fddb160000 4 4 4 rw--- libmp3lame.so.0.0.0
0000fddb161000 64 0 0 rw--- [ anon ]
0000fddb180000 320 64 0 r-x-- libmpg123.so.0.49.4
0000fddb18d000 60 0 0 ---- libmpg123.so.0.49.4
0000fddb18df000 4 4 4 r---- libmpg123.so.0.49.4
0000fddb18e000 4 4 4 rw--- libmpg123.so.0.49.4
0000fddb190000 4076 64 0 r-x-- libopus.so.0.11.1
0000fddb19feb000 72 0 0 ---- libopus.so.0.11.1
0000fddb19ffd000 12 12 12 r---- libopus.so.0.11.1
0000fddb1a0000 4 4 4 rw--- libopus.so.0.11.1
0000fddb1a00000 556 128 0 r-x-- libvorbisenc.so.2.0.12
0000fddb1a09b000 56 0 0 ---- libvorbisenc.so.2.0.12
0000fddb1a09b000 92 92 92 r---- libvorbisenc.so.2.0.12
0000fddb1a0c0000 4 4 4 rw--- libvorbisenc.so.2.0.12
0000fddb1a0d0000 304 64 0 r-x-- libFLAC.so.14.0.0
0000fddb1a11c000 76 0 0 ---- libFLAC.so.14.0.0
0000fddb1a12f000 4 4 4 r---- libFLAC.so.14.0.0
0000fddb1a130000 4 4 4 rw--- libFLAC.so.14.0.0
0000fddb1a40000 80 64 0 r-x-- libapparmor.so.1.29.4
0000fddb1a54000 108 0 0 ---- libapparmor.so.1.29.4
0000fddb1a6f000 4 4 4 r---- libapparmor.so.1.29.4
0000fddb1a70000 4 4 4 rw--- libapparmor.so.1.29.4
0000fddb1a80000 20 20 0 r-x-- libasyncns.so.0.3.1
0000fddb1a85000 104 0 0 ---- libasyncns.so.0.3.1
0000fddb1a9f000 4 4 4 r---- libasyncns.so.0.3.1
0000fddb1aa0000 4 4 4 rw--- libasyncns.so.0.3.1
0000fddb1ab0000 524 64 0 r-x-- libsndfile.so.1.0.37
0000fddb1a233000 104 0 0 ---- libsndfile.so.1.0.37
0000fddb1a24d000 12 12 12 r---- libsndfile.so.1.0.37
0000fddb1a250000 4 4 4 rw--- libsndfile.so.1.0.37
0000fddb1a251000 8 0 0 rw--- [ anon ]
0000fddb1a260000 488 448 0 r-x-- libpulsecommon-17.0.so
0000fddb1a2da000 80 0 0 ---- libpulsecommon-17.0.so
0000fddb1a2ee000 0 0 0 r---- libpulsecommon-17.0.so
0000fddb1a2f0000 4 4 4 rw--- libpulsecommon-17.0.so
0000fddb1a300000 304 256 0 r-x-- libpulse.so.0.24.3
0000fddb1a34c000 72 0 0 ---- libpulse.so.0.24.3

0000fde55afd000 12 12 12 r---- libgobject-2.0.so.0.8800.0
0000fde55b00000 4 4 4 rw--- libgobject-2.0.so.0.8800.0
0000fde55b01000 4 4 0 r---- LC_NUMERIC
0000fde55b02000 56 56 56 rw--- [ anon ]
0000fde55b10000 2016 1728 0 r-x-- libgio-2.0.so.0.8800.0
0000fde55d00000 68 0 0 ---- libgio-2.0.so.0.8800.0
0000fde55d19000 28 28 28 r---- libgio-2.0.so.0.8800.0
0000fde55d20000 4 4 4 rw--- libgio-2.0.so.0.8800.0
0000fde55d21000 0 0 0 rw--- [ anon ]
0000fde55d23000 4 4 0 r---- LC_TIME
0000fde55d24000 40 40 40 rw--- [ anon ]
0000fde55d2e000 4 4 0 r---- LC_COLLATE
0000fde55d2f000 4 4 0 r---- LC_MONETARY
0000fde55d30000 4 4 0 r---- SYS_LC_MESSAGES
0000fde55d31000 4 4 0 r---- LC_PAPER
0000fde55d32000 4 4 0 r---- LC_NAME
0000fde55d33000 4 4 0 r---- LC_ADDRESS
0000fde55d34000 4 4 0 r---- LC_TELEPHONE
0000fde55d35000 4 4 0 r---- LC_MEASUREMENT
0000fde55d36000 28 28 0 r--s- gconv-modules.cache
0000fde55d3d000 4 4 0 r---- LC_IDENTIFICATION
0000fde55d3e000 4 4 4 rw--- [ anon ]
0000fde55d3f000 4 4 4 rw--- [ anon ]
0000fde55d40000 3028 3008 0 r-x-- libshell-18.so
0000fde56035000 96 0 0 ---- libshell-18.so
0000fde5604d000 12 12 12 r---- libshell-18.so
0000fde56050000 4 4 4 rw--- libshell-18.so
0000fde56051000 24 24 24 rw--- [ anon ]
0000fde56057000 168 164 0 r-x-- ld-linux-aarch64.so.1
0000fde56081000 56 56 56 rw--- [ anon ]
0000fde5608f000 16 0 0 r---- [ anon ]
0000fde56093000 8 8 0 r-x-- [ anon ]
0000fde56095000 8 8 8 r---- ld-linux-aarch64.so.1
0000fde56097000 4 4 4 rw--- ld-linux-aarch64.so.1
0000fde56098000 4 4 4 rw--- [ anon ]
0000fffc1b2d000 132 92 92 rw--- [ stack ]
-----
total kB 5081776 472208 324548
ubuntu@ubuntu:~$

```